

AP CSP | Lesson 16: Study Goal Tracker

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B: the remainder is 2 and the product is 8, giving 10. A groups the addition before multiplication; C groups $5+2$ as the divisor; D uses division instead of MOD.

Q2. Answer: C

C is true AND true. A contains false $x < 4$; B negates true; D combines two false comparisons.

Q3. Answer: D

D includes the equality boundary. A treats $\geq$ as $>$; B ignores exclusive branch choice; C ignores that an IF/ELSE always chooses a branch.

Q4. Answer: A

A: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8$. B adds or multiplies by the count; C performs two updates; D performs four.

Q5. Answer: B and D

B follows the pre-check rule. D describes an unreachable stopping condition. A reverses true/false behavior; C does not show progress toward the required condition. Both B and D are required.

Q6. Answer: C

C counts outcomes 1 and 2 among five. A assumes two equal categories; B counts one outcome; D counts the failures.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 5. (b) 10. (c) 2. (d) AP CSP, with one space from the second string.

1 mark: Gives $y = 5$ after line 2.

1 mark: Gives $x = 10$ after line 3.

1 mark: Gives final $y = 2$.

1 mark: Gives AP CSP with exactly one space from the second string.

Q8. 4 marks

Answer: (a) Guest. (b) Junior. (c) Member. (d) The outer age condition is false, so its nested member test is skipped.

1 mark: Gives Guest for 12/false.

1 mark: Gives Junior for 11/true.

1 mark: Gives Member for 12/true.

1 mark: Explains that the false outer condition skips the nested test.

Q9. 4 marks

Answer: Final $n = 8$; three body runs. Equality to 7 never occurs because n stays even. Any initial $n \geq 7$ gives zero runs, such as 7.

1 mark: Gives final $n = 8$.

1 mark: Gives three body executions.

1 mark: Explains that equality to odd 7 is unreachable from even values.

1 mark: Gives a starting value ≥ 7 and states zero body runs.

Q10. 4 marks

Answer: Theoretical probability $1/2$; observed $8/20 = 0.4$. No: the next trial still has probability $1/2$. A real coin or tossing method may be biased or trials may not be independent.

1 mark: Gives theoretical probability $1/2$.

1 mark: Calculates observed frequency 0.4 .

1 mark: Rejects a forced next result and refers to the unchanged independent probability.

1 mark: Gives a relevant real-world assumption that may fail.

Q11. 4 marks

Answer: (a) 2, On track. (b) 0, Try again. (c) 3, On track. (d) They lie on opposite sides of the included goal boundary and distinguish >30 from ≥ 30 .

1 mark: Gives 2 and On track for 29/30/31.

1 mark: Gives 0 and Try again for 0/0/0.

1 mark: Gives 3 and On track for 30/30/30.

1 mark: Explains how 29/30 test the inclusion boundary.

Q12. 4 marks

Answer: Faulty count 1. Move count $\leftarrow 0$ before REPEAT 3 TIMES and keep the conditional increment inside. Correct result: 3, On track. Record the reset bug, moved initialization and 30/30/30 retest changing 1 to 3.

1 mark: Gives faulty count 1.

1 mark: Places count $\leftarrow 0$ before the loop and conditional increment inside.

1 mark: Gives corrected count 3 and On track.

1 mark: Records the specific change and the 30/30/30 before/after test.